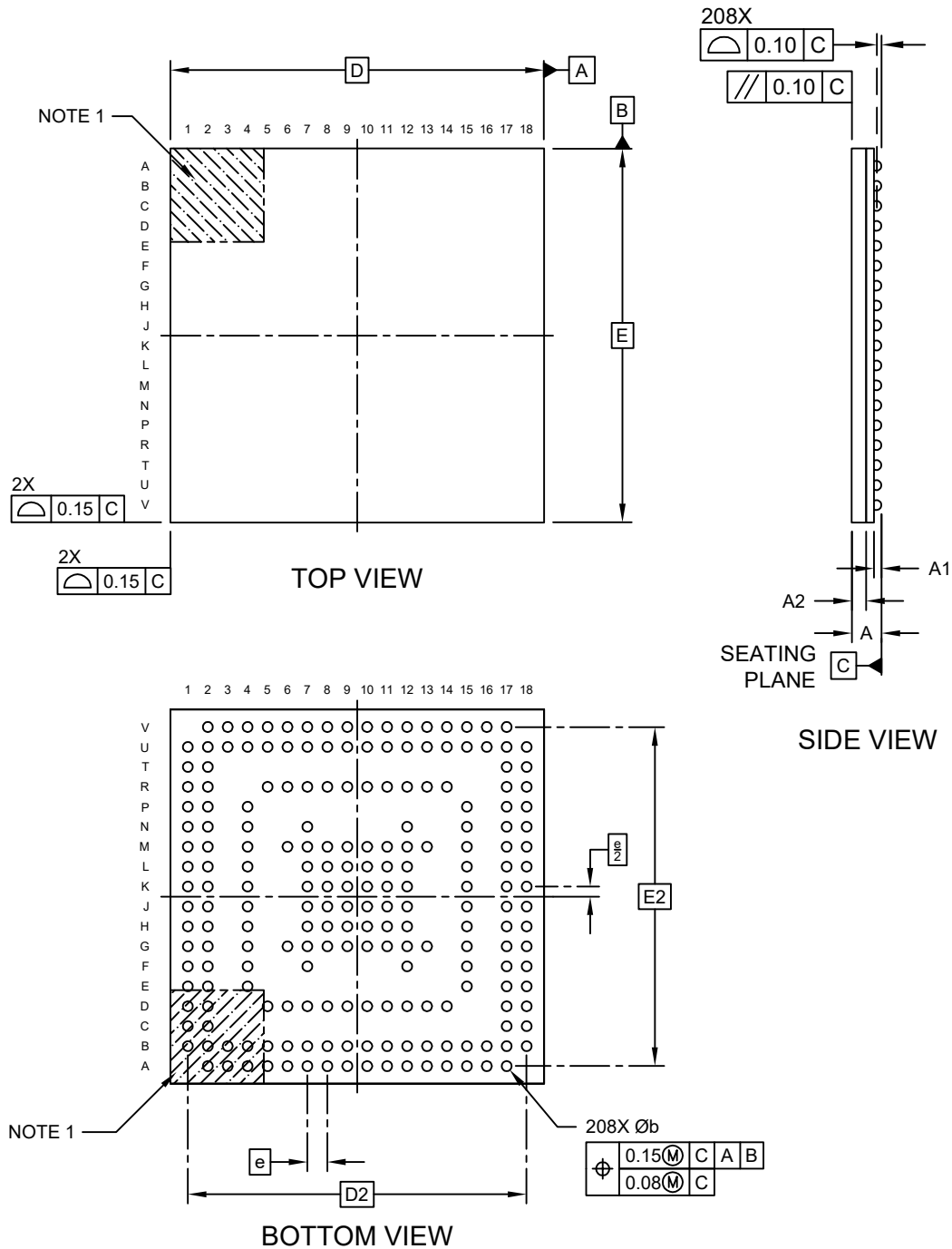


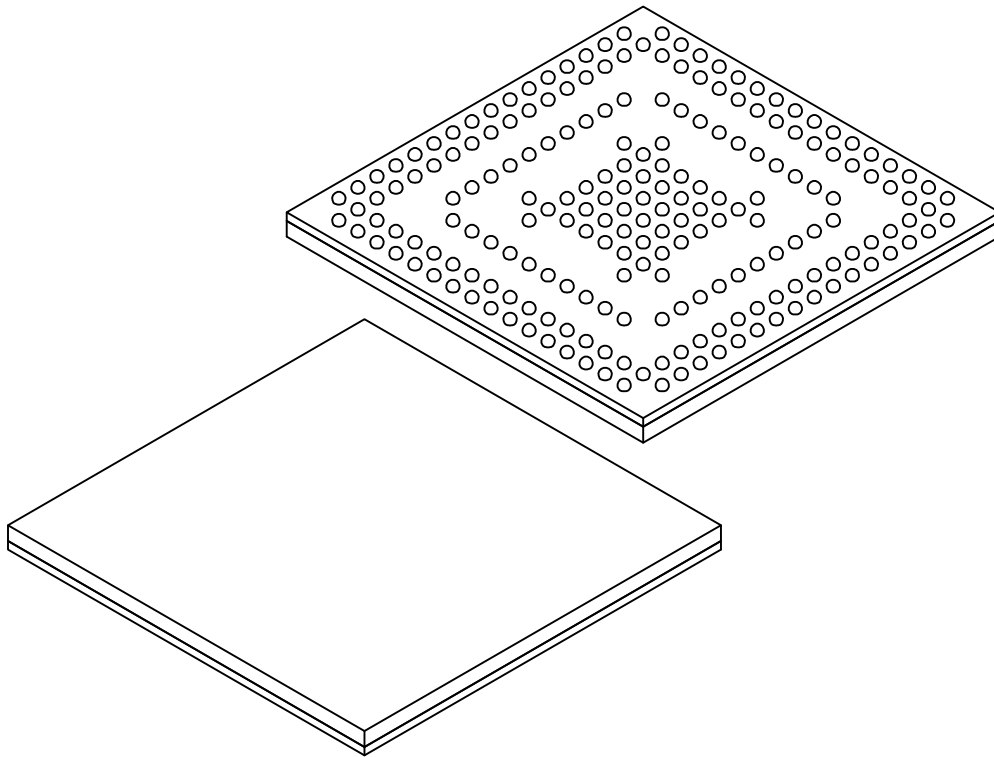
## 208-Ball Fine-Pitch Ball Grid Array Package (8MX) - 15x15x1.19 mm Body [TFBGA]

**Note:** For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



## 208-Ball Fine-Pitch Ball Grid Array Package (8MX) - 15x15x1.19 mm Body [TFBGA]

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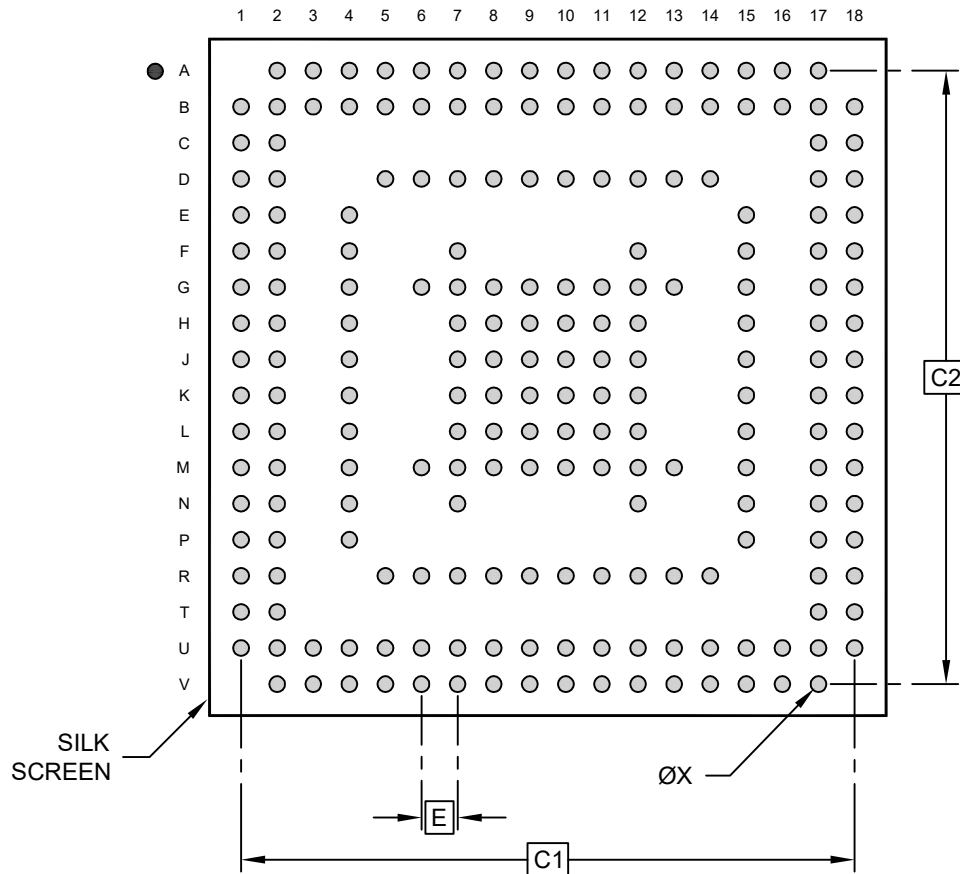
| Units               |    | MILLIMETERS |      |      |
|---------------------|----|-------------|------|------|
| Dimension Limits    |    | MIN         | NOM  | MAX  |
| Number of Terminals | N  | 208         |      |      |
| Pitch               | e  | 0.80 BSC    |      |      |
| Overall Height      | A  | –           | –    | 1.19 |
| Ball Height         | A1 | 0.21        | 0.30 | –    |
| Mold Thickness      | A2 | 0.48        | 0.53 | 0.58 |
| Overall Length      | D  | 15.00 BSC   |      |      |
| Ball Array Length   | D2 | 13.60 BSC   |      |      |
| Overall Width       | E  | 15.00 BSC   |      |      |
| Ball Array Width    | E2 | 13.60 BSC   |      |      |
| Ball Diameter       | b  | 0.35        | 0.40 | 0.45 |

### Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.  
REF: Reference Dimension, usually without tolerance, for information purposes only.

## 208-Ball Fine-Pitch Ball Grid Array Package (8MX) - 15x15x1.19 mm Body [TFBGA]

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RECOMMENDED LAND PATTERN

| Units                |    | MILLIMETERS |     |      |
|----------------------|----|-------------|-----|------|
| Dimension Limits     |    | MIN         | NOM | MAX  |
| Contact Pitch        | E  | 0.80 BSC    |     |      |
| Contact Pad Spacing  | C1 | 13.60 BSC   |     |      |
| Contact Pad Spacing  | C2 | 13.60 BSC   |     |      |
| Contact Pad Diameter | X  |             |     | 0.35 |

### Notes:

- Dimensioning and tolerancing per ASME Y14.5M  
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
- For best soldering results, thermal vias, if used, should be filled or tented to avoid solder loss during reflow process

Microchip Technology Drawing C04-2516 Rev A